

# SU(3) Flavor

Mesons, baryons w/ u, d, s

$$Q = +2/3 \quad \begin{pmatrix} u \\ d \end{pmatrix} \quad \begin{pmatrix} c \\ s \end{pmatrix} \quad \begin{pmatrix} t \\ b \end{pmatrix}$$

$$Q = -1/3$$

Meson:  $q_i \bar{q}_j$

Baryon:  $q_1 q_2 q_3$

$$q_i = u, d, s, c, b$$

Can we have  $u\bar{c}$ ,  $c\bar{s}$  ?

$SU(3)_F$ : good symm. but approximate / lightly broken

if exact symmetry



all multiplets

should be

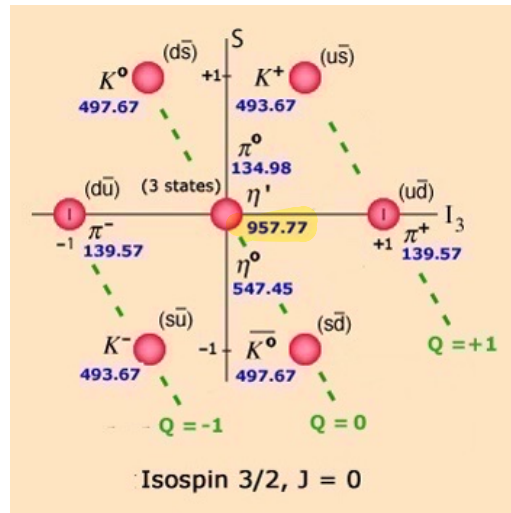
degenerate in mass

$\sim 500 \text{ MeV}$

$\sim 135 \text{ MeV}$

$\sim 1 \text{ GeV}$

$\sim 500 \text{ MeV}$



$$\Delta S = 2$$

$\Rightarrow$  Change in mass



$$m_u \approx m_d$$

$$m_{u,d} \neq m_s$$

| Baryons                                     | qqq                | $J^P$           | $I$           | $I_3$                                                  | $S$ | $Q^{(1)}$   | mass (MeV) |
|---------------------------------------------|--------------------|-----------------|---------------|--------------------------------------------------------|-----|-------------|------------|
| p, n                                        | uud, udd           | $\frac{1}{2}^+$ | $\frac{1}{2}$ | $\frac{1}{2}, -\frac{1}{2}$                            | 0   | 1, 0        | 940        |
| $\Lambda$                                   | uds                | $\frac{1}{2}^+$ | 0             | 0                                                      | -1  | 0           | 1115       |
| $\Sigma^+, \Sigma^0, \Sigma^-$              | uus, uds, dds      | $\frac{1}{2}^+$ | 1             | 1, 0, -1                                               | -1  | 1, 0, -1    | 1190       |
| $\Xi^0, \Xi^-$                              | uss, dss           | $\frac{1}{2}^+$ | $\frac{1}{2}$ | $\frac{1}{2}, -\frac{1}{2}$                            | -2  | 1, 0        | 1320       |
| $\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$ | uuu, uud, udd, ddd | $\frac{3}{2}^+$ | $\frac{3}{2}$ | $\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$ | 0   | 2, 1, 0, -1 | 1230       |
| $\Sigma^{*+}, \Sigma^{*0}, \Sigma^{*-}$     | uus, uds, dds      | $\frac{3}{2}^+$ | 1             | 1, 0, -1                                               | -1  | 1, 0, -1    | 1385       |
| $\Xi^{*0}, \Xi^{*-}$                        | uss, dss           | $\frac{3}{2}^+$ | $\frac{1}{2}$ | $\frac{1}{2}, -\frac{1}{2}$                            | -2  | 1, 0        | 1530       |
| $\Omega^-$                                  | sss                | $\frac{3}{2}^+$ | 0             | 0                                                      | -3  | -1          | 1670       |

what about the other quarks?

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix}$$

$$|u\rangle = |I=1/2, I_3=+1/2\rangle$$

$$|d\rangle = |I=1/2, I_3=-1/2\rangle$$

$$|s\rangle = |I=0, I_3=0\rangle \quad \text{strangeness} = \phi.$$

$$|c\rangle \equiv I_3 = \phi, S = \phi, C = +1$$

What about  $SU(4)$  Flavor symmetry?

$SU(4)$ : 16-1 generators  
4-1 diagonal generators.

$\left. \begin{array}{l} I_3 \\ \text{strangeness} \\ \text{charm quantum number} \end{array} \right\}$  diagonal generators

$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad d = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad s = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad c = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Gell-Mann

$$Q = I_3 + \frac{Bary + S + C}{2}$$

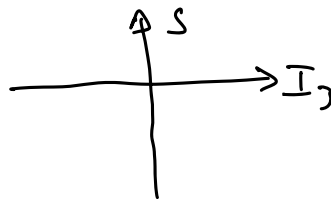
Nishijima

$$Meson = q_i \bar{q}_j \quad i = u, d, s, c$$

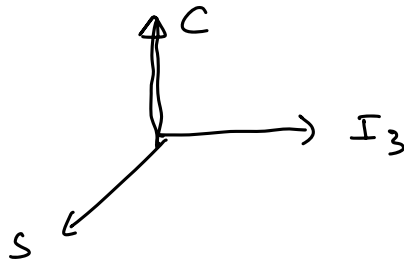
$$Meson = 4 \otimes \bar{4} = 11 + 15$$

$4 \equiv$  fundamental rep.

$SU(3)$



$SU(4)$



$$\begin{pmatrix} u \\ d \\ s \\ c \end{pmatrix}$$

$$\bar{4} \equiv \begin{pmatrix} \bar{u} \\ \bar{d} \\ \bar{s} \\ \bar{c} \end{pmatrix}$$

$$4 \times \bar{4} = 11 + 8 + 6$$

D mesons:

$$\begin{array}{cc} c & \bar{u} & D^0 \\ & \bar{d} & D^+ \\ & \bar{s} & D_s^0 \end{array}$$
$$\begin{array}{lll} c \bar{c} & 1c & S=0 \\ c \bar{c} & J/\psi & S=1 \end{array}$$

$S = 0$  pseudo-scalar  $\uparrow \downarrow$   
charmed mesons

$S = 1$  vector charmed mesons  $\uparrow\uparrow \downarrow\downarrow$

$$m_{D^0, D^+} \approx 1.879 \text{ GeV}$$
$$m_D \gg m_{\text{u,d,s}} \text{ mesons}$$

$\eta_C$  J/f can be  $\mathbb{Q}$  eigenstate.

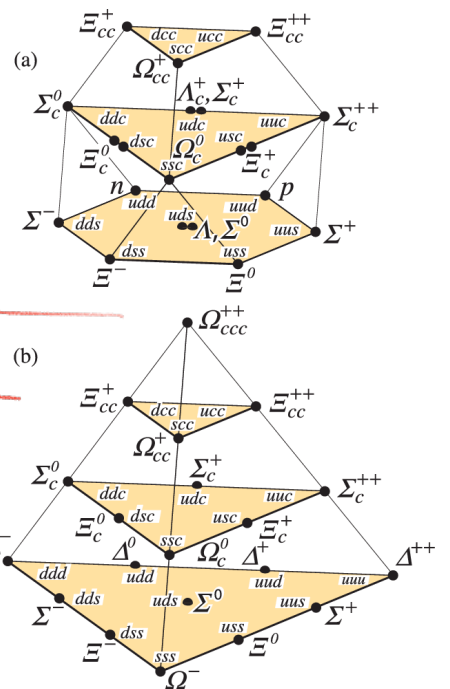
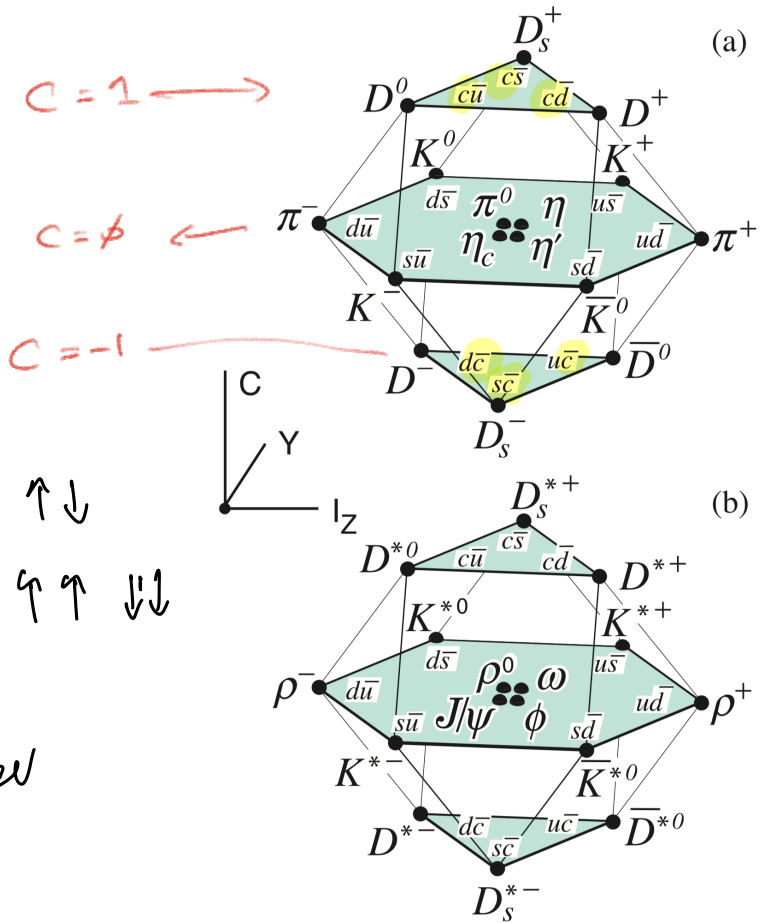
$$m_0 \neq m_k, m_{\bar{u}}, m_e \Rightarrow SU(4) \text{ is broken.}$$
$$m_{u,d} \neq m_s \neq m_c$$

$D^0$  ,  $D^{*0}$  :  $c\bar{u}$  same quark content  
 $\downarrow$   $\downarrow$   
 $S=0$   $S=1$

Charmed baryons:

$$4 \otimes 4 \otimes 4 = \mathbb{A}$$

mass change significantly  
with additional C quarks.

$$\Rightarrow m_c \gg m_s \gg m_{u,d}$$


Beauty/bottom quark:  $b$

$$S = \begin{pmatrix} u \\ d \\ s \\ c \\ b \end{pmatrix}$$

$SU(5)$  Flavor  $\Rightarrow$  4 diag. gener.

Diagonal generators:

$T_3$

Strangeness

Charm quark number

beauty quantum number

$$\Rightarrow Q = T_3 + \frac{Baryon + S + C + b}{2}$$

$$\begin{pmatrix} u \\ d \end{pmatrix} \quad \begin{pmatrix} c \\ s \end{pmatrix} \quad \begin{pmatrix} t \\ b \end{pmatrix} \rightarrow \begin{array}{l} \text{up-like quarks } Q = +\frac{2}{3} \\ \text{down-like quark } Q = -\frac{1}{3} \end{array}$$

quantum numbers  $+1$ : for uplike.

$-1$ : for down like

strange quark:  $S = -1$

charm quark:  $C = +1$

beauty quark:  $b = -1$

B mesons

$b\bar{u}$ :  $B^-$

$b\bar{d}$ :  $\bar{B}^0$

$\bar{b}u$ :  $B^+$

$\bar{b}d$ :  $B^0$

} Mass  $\sim 5.279$  GeV  
 $\gg m_D$   
 $\Downarrow$

$$m_b \gg m_c \gg m_s \gg m_{u,d}$$

multiplet of  $SU(3)$   $\Rightarrow$  4-dimensional variety/surface

$\Rightarrow$  no hadrons with  $t$ -quarks.

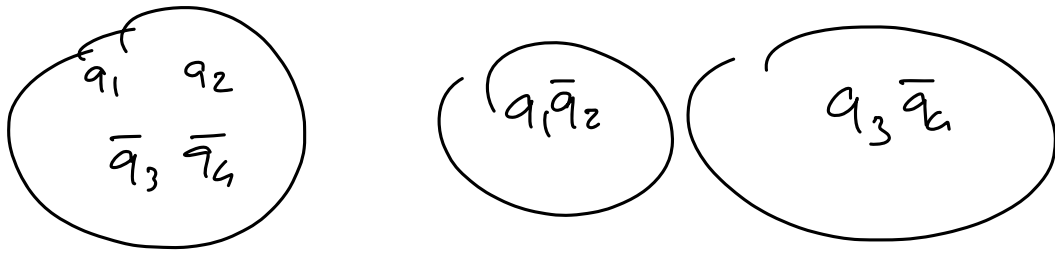
$$m_t = 172 \text{ GeV}$$

top quark discovered 1994

~~$t\bar{u}$~~ ,  ~~$t\bar{d}$~~

maybe hint for  $t\bar{t}$  bound state.  
(toponium)

tetra quarks:  $q_1 q_2 \bar{q}_3 \bar{q}_4$  in  $SU(3)$  color

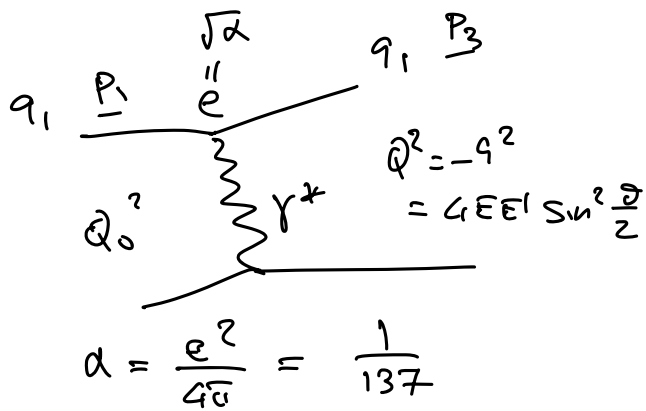


tetra quarks do not fit in  $SU(N)$  Fermi multiplets.  
seen so far

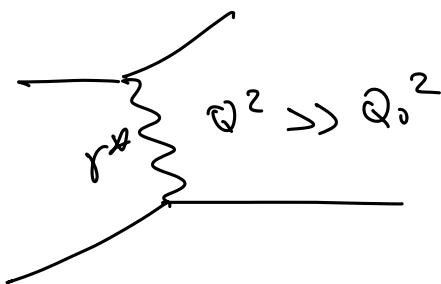
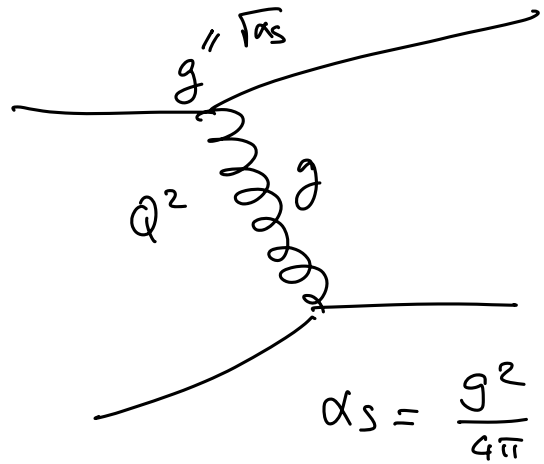
pentaquarks: hadron w/ 5 colored quarks.  
colorless.

QED

$$q = p_1 - p_3$$

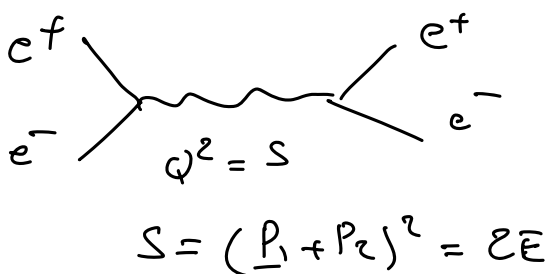


QCD

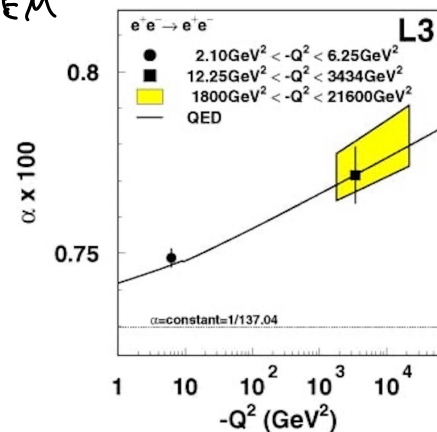


$$Q^2 = (90 \text{ GeV})^2 \Rightarrow \alpha(Q^2) = \frac{1}{12.9}$$

$$\alpha(Q^2) = \frac{\alpha(Q_0^2)}{1 - \Delta\alpha(Q^2)}$$



$\alpha_{EM}$



In QCD 
$$\alpha_s(Q^2) = \frac{\alpha_s(Q_0^2)}{1 + \frac{7}{4\pi} \alpha_s(Q_0^2) \log \frac{Q^2}{Q_0^2}}$$

